Brendan Ryan

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EDUCATION

University of Rochester Rochester, NY
Bachelor of Science, Optical Engineering *Anticipated May 2028*

- **GPA:** 4.00/4.00 | Dean's List (4/4 semesters)
- **Phi Beta Kappa Suzanne J. O'Brien Book Award:** Selected as 1 of 14 recipients among 1,179 first-year students in recognition of outstanding first-year scholarly achievement, cocurricular involvement, and leadership potential
- **Relevant Coursework:** Geometrical Optics, Physical Optics, Aberrations, Interferometry & Metrology, Optical Fabrication & Testing, MATLAB for Optics I & II
- **Spring Coursework:** Lens Design, Optomechanical Design, Sources & Detectors

EXPERIENCE

Optimax Systems, Inc. Ontario, NY
Optical Systems Intern *August 2026 – Present*

- Assemble and align precision optical systems within ISO 5 cleanroom and ESD-controlled environments, applying active and passive alignment techniques
- Verify optical system performance using transmitted wavefront interferometry and polarization purity testing
- Design CAD models using SolidWorks and apply GD&T to develop custom tooling, test coupons, and a transmission sphere assembly supporting optical assembly workflows

Stealth AI Startup Bellevue, WA
Systems Test Engineer Intern *June 2026 – July 2026*

- Designed and executed hardware and system-level evaluations to validate performance under real-world conditions
- Documented methods and results and presented them in a technical briefing, translating test data into clear engineering recommendations
- Conducted hardware research and competitive analysis to support technical and business decisions

University of Rochester, Prof. Gerald Gamm Rochester, NY
Independent Study Researcher *September 2025 – Present*

- Compile and process primary-source data in Excel for quantitative analysis in an academic research paper
- Track and analyze legislative actions on reproductive-rights bills using historical journals from multiple states

Technical Glass Products Snoqualmie, WA
Powder Coat Processor *May 2025 – August 2025*

- Performed quality-control inspections to verify coating integrity on fire-rated frames, doors, and windows
- Followed SOPs for safe handling and processing in a high-heat production environment

PROJECTS

Kestrova https://github.com/br24563/Kestrova
Optical Design & Analysis Software, Personal Project *August 2026 – Present*

- Developing optical design software for ray tracing, aberration analysis, and optical tolerancing

GlassMatch https://github.com/br24563/Glass-Match
Optical Glass Selection Database, Personal Project *September 2026 – Present*

- Developing an open-source database to compare optical glasses and identify equivalent materials across manufacturers

Telecentric Stereo Microscope Design Rochester, NY
Geometrical Optics Course, University of Rochester *November 2025 – December 2025*

- Designed a first-order telecentric stereo microscope with an integrated camera path for art conservation applications
- Balanced magnification, working distance, and resolution to realize the design using catalog lenses

CAMPUS LEADERSHIP & ACTIVITIES

NCAA Division III Varsity Baseball Rochester, NY
Analytics Assistant (October 2026 – Present); Pitcher (August 2024 – October 2026) *August 2024 – Present*

- Support the coaching staff with player development by analyzing Trackman pitch-tracking data and video
- Competed as a pitcher for two seasons while balancing a full optical engineering course load with year-round training
- Named to the 2026 Liberty League All-Academic Team for academic excellence as a student-athlete

SKILLS

- **Optics:** Ray tracing, optical alignment, encircled energy testing, Zygo Verifire, ZeGage, Optikos LensCheck
- **Design & Software:** SolidWorks, GD&T, tooling design, MATLAB, Microsoft Excel, PowerPoint, ImageJ
- **Engineering:** Test planning, experimental design, technical writing, presentations, data analysis